

STREAMING EVENT ANOMALY DETECTOR

TEST CASES & SAMPLE DATA

INTRODUCTION

This document contains functional, security, performance, and AI validation test cases for the Streaming Event Anomaly Detector. These test cases can be used during development, demonstrations, interviews, project reviews, and final evaluations.

TEST CASE SUMMARY

Test Case ID	Test Case Name	Module
TC-01	Normal Event Processing	Event Processing
TC-02	High Value Anomaly Detection	Anomaly Engine
TC-03	Low Value Anomaly Detection	Anomaly Engine
TC-04	AI Explanation Generation	AI Module
TC-05	AI Agent Severity Classification	AI Agent
TC-06	Discord Alert Notification	Alert Service
TC-07	User Login Validation	Authentication
TC-08	Invalid Login Attempt	Authentication
TC-09	Dashboard Data Refresh	Dashboard
TC-10	Event History Storage	Database
TC-11	High Volume Event Load	Performance
TC-12	Database Recovery Validation	Reliability

SAMPLE EVENT DATA

Normal Event

Input

{

```
"event_id": 1001,  
"source": "server_01",  
"value": 120  
}
```

Expected Result

- Event accepted
- Stored in database
- No anomaly generated

Critical Event

Input

```
{  
  "event_id": 1002,  
  "source": "server_01",  
  "value": 5000  
}
```

Expected Result

- Event accepted
- Anomaly detected
- AI analysis generated
- Alert triggered

Low Event

Input

```
{  
  "event_id": 1003,  
  "source": "server_01",  
  "value": -1000  
}
```

Expected Result

- Event accepted
- Negative anomaly detected
- Severity assigned
- Alert generated if required

TC-01

Normal Event Processing

Input

```
{  
  "value": 150  
}
```

Expected Result

- Event processed successfully
- No anomaly generated
- Dashboard updated

Status

PASS

TC-02

High Value Anomaly Detection

Input

```
{  
  "value": 5000  
}
```

Expected Result

- Z-Score exceeds threshold
- Critical anomaly detected

Status

PASS

TC-03

Low Value Anomaly Detection

Input

```
{  
  "value": -2000  
}
```

Expected Result

- Negative anomaly identified
- AI explanation generated

Status

PASS

TC-04

AI Explanation Generation

Scenario

An anomaly has been detected.

Expected Result

AI generates explanation similar to:

"This event significantly deviates from the historical average and indicates abnormal system behavior."

Status

PASS

TC-05

AI Agent Severity Classification

Input

Z-Score = 4.8

Expected Result

Severity:

CRITICAL

Recommended Action:

Immediate investigation required.

Status

PASS

TC-06**Discord Alert Notification****Scenario**

Critical anomaly detected.

Expected Result

- Discord webhook called
- Alert delivered successfully

Example Message

CRITICAL ALERT

Event ID: 1002

Severity: CRITICAL

Recommendation: Immediate Investigation

Status

PASS

TC-07**User Login Validation****Input**

```
{  
  "username": "admin",
```

```
"password":"admin123"  
}
```

Expected Result

- JWT token generated
- User redirected to dashboard

Status

PASS

TC-08**Invalid Login Attempt****Input**

```
{  
  "username":"admin",  
  "password":"wrongpassword"  
}
```

Expected Result

HTTP 401 Unauthorized

Status

PASS

TC-09**Dashboard Data Refresh****Scenario**

New anomaly enters system.

Expected Result

- Dashboard refreshes automatically
- Metrics updated

Status

PASS

TC-10

Event History Storage

Scenario

Event processing completed.

Expected Result

- Event stored in database
- Timestamp recorded
- Event retrievable

Status

PASS

TC-11

High Volume Event Load Test

Scenario

Generate 10,000 events.

Expected Result

- System remains stable
- Events processed continuously
- No data loss

Performance Target

- Success Rate > 95%
- Processing Delay < 2 Seconds

Status

PASS

TC-12

Database Recovery Validation

Scenario

System restart occurs.

Expected Result

- Historical data preserved
- Previous anomalies accessible
- Users retained

Status

PASS

SECURITY TEST CASES**STC-01 JWT Validation**

Expected Result

- Unauthorized requests blocked

STC-02 Password Hash Verification

Expected Result

- Passwords stored as hashes
- Plain text passwords never stored

STC-03 Protected Endpoint Access

Expected Result

- Only authenticated users can access APIs

PERFORMANCE TEST CASES**PTC-01 Concurrent Event Processing**

Input

1000 simultaneous events

Expected Result

System processes all events successfully.

PTC-02 Dashboard Performance

Expected Result

Dashboard loads within 3 seconds.

PTC-03 AI Response Time

Expected Result

AI explanation generated within 10 seconds.

ACCEPTANCE CRITERIA

- ✓ Event processing functional
- ✓ Anomaly detection operational
- ✓ AI explanations generated
- ✓ Severity classification working
- ✓ Discord alerts operational
- ✓ Dashboard updates in real time
- ✓ Authentication secured
- ✓ Database persistence verified
- ✓ Performance targets achieved

CONCLUSION

The Streaming Event Anomaly Detector has been validated through functional, security, AI, and performance testing. All major modules including anomaly detection, AI analysis, alerting, authentication, dashboard monitoring, and database persistence operate successfully and satisfy project requirements.